Wiren Board 7.4 — cheat sheet

COMPETITION MASTER 03 — Wiren Board 7.4 Complete Smart Farm

Connections to remember

Controller pin	Goes to
A1 (A1)	pH · AO (analog)
A2 (A2)	EC · AO (analog)
A3 (A3)	Water level · AO (analog)
A4 (A4)	ACS712 pump current · OUT
GND	12 V 10 A · –
GND	5 V 1 A · –
RS-485-1 A	WB-MR6C (addr 32) · A
RS-485-1 A (RS-485 A)	Air sensor RS-485 (addr 2) · A+ (yellow)
RS-485-1 B	WB-MR6C (addr 32) · B
RS-485-1 B (RS-485 B)	Air sensor RS-485 (addr 2) · B− (blue)
V+	12 V 10 A · +

Settings

```
MAX_TEMP = 30  ·  FAN_OFF_TEMP = 27  ·  MIN_WATER = 20  ·  PUMP_SECONDS = 5  ·
IRRIGATION_HOURS = [7, 12, 17]  ·  LIGHT_ON_HOUR = 6  ·  LIGHT_OFF_HOUR = 20  ·  DOSE_A_ML =
10  ·  DOSE_B_ML = 5  ·  FLOW_ML_PER_SEC = 2  ·  PH_MIN = 5.5  ·  PH_MAX = 6.5  ·  EC_MIN
= 1.0  ·  EC_MAX = 2.5  ·  MIN_PUMP_AMPS = 0.2  ·  TEST_SECONDS = 2
```

Key lines of the program

wb/rules.js

```
var TEMP = "air-sensor_2/Temperature";
var PH_VOLTS = "wb-adc/A1";
var PUMP = "wb-mr6c_32/K1";
defineVirtualDevice("farm", {
  if (getControl(name).getError() !== "") return null;
  if (dev[name] !== on) dev[name] = on;
  setTimeout(doseBStart, DOSE_A_ML / FLOW_ML_PER_SEC * 1000);
  var hour = new Date().getHours();
  whenChanged: "farm/start_dosing",
  whenChanged: "farm/start_test",
setInterval(control, 1000);
```

The program in 6 steps

- Read the sensors (None/null when one fails).
- Buttons choose the mode: AUTO, DOSING, TEST.
- AUTO: fan 30/27 °C, light 06–20 h, irrigation at 7/12/17 h if water ≥ 20 %.
- DOSING: water check, B off, A for 5 s, A off, B for 2.5 s, B off.
- Alarms: SENSOR ERROR → LOW WATER → PUMP FAILURE → HIGH TEMPERATURE → OUT OF RANGE.
- Status every 5 s, wait 1 s, repeat.